Atomic nuclei
discovery of, 7
neutron approach to, 128, 129f
nuclear radiation absorption and, 165t
with nucleon spin–orbit coupling, 139–141, 140t
without nucleon spin–orbit coupling, 138–139, 138t
Schrödinger wave equation for, 129–130
species of, 32–33
stability of, 65–84
binding energy in, 72–74, 73f
mass defect, 69–72, 71t
missing elements, 79–82
neutron to proton ratio in, 67–69, 69f
nuclear radius in, 74–75, 75f
other modes of instability, 82
patterns of, 65–67, 66f
promethium, 79, 80f
semiempirical mass equation, 75–77
technetium, 80–82, 81f
valley of β-stability, 77–79, 77f, 78f
Atomic number (Z), 32
mass number relationship to, 32
notation for, 6
in radioactive decay, 6–7
Atomic reaction cross-section, 182–183
Atomic weights (M), 34
ATR, *see* Advanced thermal reactor
Attenuation coefficient, *see* Total attenuation coefficient
Attractive nuclear force, 300–301, 301f
Auger electrons, 106, 186
Autobatch procedures, 535–537, 536f, 537f
Autoradiography, 573
in life sciences, 575, 576f
Availability factor, 631
Azimuthal quantum number (l), 129–130
nuclear energy levels with, 138t

B

B, *see* Dose build-up factor
Back scattering, 180–182
absolute disintegration rates and, 286
of beta particles, 180–182, 181f
of gamma rays, 280–281, 280f
Background eradication, 242, 290
Backing material, of target material, 523
Baryons
mass classification of, 20
properties of, 21t, 26
substructure of, 17
Bateman equations, 520
Becquerel (Bq), 112
Beginning of time, 350–352, 352f
Berkelium (Bk), 423–424
Berthelot-Nernst homogeneous distribution law, 551
Beryllium (Be)
isotopic data for, 34t
neutron moderation with, 606t
Beta decay, 4, 68, 86–87, 90–97, *see also* Electron capture, Electron emission, Positron emission
daughter recoil, 97
decay energy of, 94
half-life and, 119
detection of, 90
double, 94–95
energy spectrum of, 91–92, 91f
fermion creation in, 22
isotopes from, 4
neutrino creation in, 23–25, 92–94, 93f
neutron to proton ratio and, 67
nuclear structure and, 154–155, 154t
process of, 91–92, 91f
water range of, 172, 172t
weak interaction force and, 18
Beta particles, 175
absorption of, 165t, 175–182, 176f
bremsstrahlung, 176–177, 176f, 177f, 178f
Čerenkov radiation, 176f, 178, 179f
excitation, 176f
ionization, 175–176, 176f, 177f
positron annihilation, 176f, 178–179
back scattering of, 180–182, 181f
Čerenkov detectors for, 274
range of, 180, 181f
scintillator detection of, liquid, 272
Beta radiography, 203, 204t
Beta spectroscopy
sample preparation for, 286
surface barrier detectors for, 264–265
Beta-current neutron detector, 257–258
Bh, *see* Bohrium

Big Bang theory, 339
elementary particles formed in, 28t
microwave radiation and, 349–350
neutrinos and, 25–26
Bimolecular reaction, energies in, 49–51, 50f
Binding energy (E_B), 72
atomic mass and, 71t
excitation energy and, 318
fission and, 74
fluorescent radiation and, 106
fusion and, 73–74
for helium, 72
in internal conversion, 100
of neutron, 128, 129f
in nuclear stability, 72–74, 73f
nucleon and, 73–74, 73f, 76
Biological affinity, of radioactive tracers, 574–575
autoradiography using, 575, 576f
DNA analysis using, 577
RIA using, 575
Biological effects of radiation, 446, *see also* Radiation biology
biological target, 448f, 449–453
on different cell types, 454f
genetic effects, 467
on molecular level, 449–450
radiation chemistry, 450t, 452–453
radiation physics, 450–452
radiation weighing factors, 453
radiation background, 469–473, 468t–469t
radiation sterilization, 454f, 454t, 466–467
Biological systems, isotope effects in
chemical equilibrium, 49
chemical kinetics, 49–52, 50f
Biosynthesis, radioactive labelling through, 528
Bismuth (Bi), nuclear radius of, 74–75, 75f
Bismuth phosphate process, 710
Bk, *see* Berkelium
Black holes, 343–344, 363
Blueshift, 344
Bohr magneton (B.m.), 134
Bohrium (Bh), 440t, 442
Boiling water reactors (BWRs), 657
coolant for, 598–599
corrosion in, 666–668
data for, 610t
efficiency of, 631
emergency cooling systems of, 636–637, 636f
event tree for, 681, 682f
general principles of, 610t, 626–630, 629f
history of, 12
load factor for, 631–632
overview of, 660–663
prevalence of, 598, 598t
radionuclide release from, 638, 639t
shim control in, 634–635
spent fuel from, 695, 696t
temperatures in, 692, 692f
Boltzmann distribution, 44
Bonding materials, 689
Boron (B), isotopic data for, 34t
Bosons, 22
Bottom quark, 28t
Bq, *see* Becquerel
Bragg peak, 174f, 175
Branched decay, 101f, 103–104, 113
Breeder reactors, 620–621, 669–674, *see also* Fast breeder reactor
thermal breeder, 621, 669–671, 670f
Breeding gain, 620
Bremsstrahlung
of beta particles, 176–177, 176f, 177f, 178f
in neutron stars, 367
Brown dwarfs, 343–344, 353
Bubble chambers, 240–241
liquid hydrogen, 15, 16f
particle moving in, 20
Burners, 619
Burnout, of target material, 524
Burn-up, 619
Cesium and, 695–696, 697f
BWRs, *see* Boiling water reactor, Boiling water reactors

C

^{14}C method, 390–393, 392f, 393f
Calcium
decay scheme for, 141–142, 142f
fission of, 73–74
nuclear radius of, 74–75, 75f
California University Cyclotrons (calutrons), 58
Californium (Cf), 423–424, 441
Calorimetric measurements, of radiation exposure, 229
calutrons, *see* California University Cyclotrons